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GRADE GUIDANCE SERIES

GRADE guidance 23: considering cost-effectiveness evidence in moving from evidence to health-related recommendations

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Abstract

Background: This is the 23rd in a series of articles describing the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach to grading the certainty of evidence and strength of recommendations for systematic reviews, health technology assessments, and clinical guideline development.

Objectives: We outline how resource utilization and cost-effectiveness analyses are integrated into health-related recommendations, using the GRADE Evidence to Decision (EtD) frameworks.

Study Design and Setting: Through iterative discussions and refinement, in-person, and online meetings, and through e-mail communication, we developed draft guidance to incorporate economic evidence in the formulation of health-related recommendations. We developed scenarios to operationalize the guidance. We presented a summary of the results to members of the GRADE Economic Evaluation Project Group.

Results: We describe how to estimate the cost of preventing (or achieving) an event to inform assessments of cost-effectiveness of alternative treatments, when there are no published economic evaluations. Evidence profiles and Summary of Findings tables based on systematic reviews of cost-effectiveness analyses can be created to provide top-level summaries of results and quality of multiple published economic evaluations. We also describe how this information could be integrated in GRADE's EtD frameworks to inform health-related recommendations. Three scenarios representing various levels of available cost-effectiveness evidence were used to illustrate the integration process.

Data availability: No original data

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Conclusion: This GRADE guidance provides practical information for presenting cost-effectiveness data and its integration in the development of health-related recommendations, using the EtD frameworks. © 2023 Elsevier Inc. All rights reserved.

Keywords: GRADE; Guidelines; Systematic reviews; Cost-effectiveness analysis; Evidence to decision frameworks; Economic evaluation

1. Introduction

This is the 23rd in a series of articles describing the Grading of Recommendations Assessment, Development and Evaluation (GRADE) guidance to grading the certainty of evidence and strength of recommendations for authors of systematic reviews and guideline developers. In the 10th article of the series, we described challenges with rating certainty of evidence (also known as quality of the evidence or confidence in an effect estimate) for resource use which is included in GRADE evidence profiles and Summary of Findings (SoF) tables [1]. The evidence profile and summary of finding table have been validated in multiple studies including randomized trials [2−5]. We also introduced health economic evaluation as a methodological framework for assessing the relative efficiency (cost-effectiveness) of alternative management strategies (interventions) [6].

An economic evaluation may be conducted either as part of an empirical study, such as a clinical trial or registry study, or using decision modeling. Trial-based economic evaluations measure the cost-effectiveness of alternative management strategies among trial participants by combining data on clinical effects, resource use, unit costs, and patient-reported outcomes (e.g., health state utilities) that are sometimes collected as part of a trial [7]. In contrast, model-based economic evaluations compare the cost-effectiveness of alternative management strategies in a specified target patient population, through mathematical modeling techniques. Typically, they do this by synthesizing clinical and economic data, gathered from a number of different research and administrative sources, and predicting incremental cost-effectiveness of the alternative strategies [8]. Trial-based and model-based economic evaluations are complementary approaches, for example, modeling can be used in extrapolating findings beyond the trial follow-up. The 15th article of the series [9], and the subsequent GRADE Working Group papers introducing the GRADE Evidence to Decision (EtD) frameworks [10−14], highlights resource use and cost-effectiveness as key criteria that may determine the direction and strength of guideline recommendations. Decision-makers may consider resource use and cost-effectiveness alongside the rest of the factors considered in the EtD frameworks (e.g., equity and acceptability) [8,9,15,16].

2. Objective

This article provides guidance on the incorporation of cost-effectiveness evidence into the GRADE EtD framework when making health-related decisions. Specifically, we present three scenarios and explain how to integrate economic evidence and how this information may help inform the direction and strength of recommendations.

3. Materials and methods

We conducted most of this work, before GRADE's creation of project groups and new processes for article approval and processes [17,18]. It was developed using the approach that GRADE pursued for earlier guidance articles, including discussion of the lead authors at GRADE meetings, with input from the GRADE working group. Through iterative discussions and refinement during in-person and online meetings and through e-mail communication, we developed draft guidance to using cost-effectiveness data in the formulation of recommendations and making decisions in 2013 and first submitted for publication in January 2019. Given changes in the leadership of the project group, substantial revisions that addressed the original peer reviewers' comments were completed from 2022 to 2023. This was followed by a new round of peer review. Further revisions based on comments and input received were made by the current authors to finalize this guidance.

We developed three key scenarios that highlight situations that guideline developers and other decision-makers often face, when they use GRADE EtDs and similar frameworks. These scenarios were chosen to reflect various levels of available cost-effectiveness evidence, ranging from none to multiple relevant published economic evaluations. Decision-makers should use the best available evidence to inform recommendations. A systematic literature would be needed to search and identify published evidence on cost-effectiveness [19]. However, often there might be no relevant evidence on cost-effectiveness at the time of guideline development. Scenario 1 represented a situation when there is no published cost-effectiveness evidence. In this situation, a simplified approach to consider cost and effectiveness may be useful. One such approach is by calculating the cost of preventing an event (COPE). Scenarios 2 and 3 dealt with the situations where there are published cost-effectiveness analyses (CEAs). When there was published CEAs, we developed economic evidence profiles that describe key findings of the CEAs including cost of treatment, incremental cost per patient, incremental effect per patient, incremental cost-effectiveness ratio (ICER), uncertainty,

What is new?

Key findings

- When using the GRADE Evidence to Decision (EtD) frameworks, the approach to integration of evidence about cost-effectiveness may vary by available evidence. When there are multiple cost-effectiveness analyses, top-level tabulated summaries of key findings and quality of the evaluations can be developed and presented as evidence profiles. In the absence of relevant cost-effectiveness evidence, the cost of preventing presents one alternative.

What this adds to what was known?

- GRADE EtD frameworks are widely used, but users are challenged about how to integrate cost-effectiveness evidence. This guidance describes practical scenarios for guideline developers and other decision-makers.

What is the implication and what should change now?

- Practical information for summarizing and presenting cost-effectiveness evidence is key to its integration in the development of recommendations.

and overall certainty rating. For each scenario, we used published reviews or guidelines developed using the GRADE approach to illustrate how to integrate cost-effectiveness evidence into the EtD frameworks. We described how to summarize cost-effectiveness evidence contained in the economic evidence profile in the relevant sections in the EtD frameworks and integrate into guideline recommendations.

4. Results

4.1. Scenario 1: no published cost-effectiveness evidence

Decision-makers should use the best available evidence to inform recommendations. However, often there might be no relevant evidence on cost-effectiveness at the time of guideline development. Therefore, guideline panels may undertake a structured discussion or a simple 'back-of-the-envelope' calculation. One approach is a limited form of considering cost-effectiveness, such as the COPE.

We refer to the scenario in which the guideline panel could consider whether it is plausible to estimate cost-effectiveness without a CEA given the evidence on safety and effectiveness of the intervention. For example, if an intervention is very low-cost and improves health (and therefore prevents the need for subsequent treatment), then it would be highly plausible that the intervention would be less costly (i.e., economically dominant).

The aforementioned situation can occur, but when a decision is not clear, then simple modeling approaches can be useful. The COPE represents the cost of preventing a single adverse outcome or achieving a single beneficial outcome [20]. COPE is the product of the number needed to treat (NNT) to prevent a single adverse event, or achieve a single additional success, multiplied by the total cost of treatment for a single individual over a specified time frame. For instance, if one needs to treat 20 patients for 2 years to prevent a single premature death and the cost of treatment for a single patient over 2 years is \$10,000, the COPE is 20 × \$10,000, or \$200,000. Evidently, COPE differs from properly conducted CEAs because it considers only the direct costs of providing the intervention. It does not include consideration of costs associated with the impact of the intervention on downstream use of resources or any other costs.

For example, when guideline developers formulated a recommendation regarding the use of pulmonary rehabilitation for patients with chronic obstructive pulmonary disease (COPD) and recent exacerbation of their disease [21], none of the trials included in the source systematic review reported estimates of resource use or a CEA associated with pulmonary rehabilitation. Table 1 shows NNTs with respect to the outcome 'mortality', derived from corresponding pooled effect size estimates reported in a trustworthy systematic review of pulmonary rehabilitation, when compared with usual care without rehabilitation in adults with COPD [21]. The median length of follow-up among the six randomized controlled trials included in estimating the pooled treatment effect for the outcome 'mortality' was 12 months. The inference from NNTs was that 23 COPD patients would need to be treated with an initial program of pulmonary rehabilitation, following a recent exacerbation of their disease, to prevent one death during a subsequent 12-month period. The estimated upfront costs of treating one patient using a program of pulmonary rehabilitation were assumed to be \$3,000 to \$5,000 per patient (for good practices on costing, please refer to Drummond et al. [22]). In this example, there is no need to annualize the costs of treatment because pulmonary rehabilitation is provided as a discrete, time-limited program. The COPEs are shown in the final column of Table 1. Estimates of the cost of preventing a single death through the use of pulmonary rehabilitation for patients with COPD (with recent exacerbation of their disease) over 12 months range between approximately \$70,000 and \$116,000 (2010 US Dollars). Of note, with this approach we are not grading the certainty of the COPE for this outcome; rather we draw upon the benefit/harm and cost outcomes, based on the evidence profile published. Such estimates help guideline developers to answer the question how large the resource requirements (costs) are in various settings. Guideline developers should

Table 1. Cost of preventing a death (COPE): pulmonary rehabilitation vs. usual care

Outcome	Risk with control	Risk with rehabilitation	NNT	Cost	COPE
Mortality	150 per 1,000	107 per 1,000	23	$3,000	$69,000
Mortality	150 per 1,000	107 per 1,000	23	$4,000	$92,000
Mortality	150 per 1,000	107 per 1,000	23	$5,000	$115,000

NNT, number needed to treat was calculated based on absolute differences in mortality at 12 mo after pulmonary rehabilitee vs. usual care in the review [21]. COPE=NNT*Cost (e.g., 23 × $3,000 = $69,000).

define, ideally, a priori thresholds of acceptable in various countries resource requirements.

The systematic review also reported a pooled estimate of the effect of pulmonary rehabilitation on the rate of hospital readmission, over a median of 9 months follow-up [21]. We calculated that five patients would need to be treated to prevent one hospital readmission [21]. One implication of this finding is that, if costs associated with avoided hospital readmissions are the only important impact of pulmonary rehabilitation on downstream use of resources, then COPE for mortality based on treatment costs would overestimate the cost of preventing a single death due to the cost saving from avoided readmissions. A further implication is that given a sufficient number of hospital readmissions avoided, cost-savings from avoided hospitalization in the future may entirely offset upfront treatment costs, and pulmonary rehabilitation would become a cost-saving management strategy compared with usual care.

It is possible to elaborate the basic form of COPE presented above by combining respective NNTs for 'mortality' and 'hospital readmission', with information about the cost of treatment and the cost of a hospital bed per day. By varying the estimated impact on length of stay for readmitted patients it would be possible to estimate the 'switching point' at which cost-savings would entirely offset upfront treatment costs. It should be noted that COPE might be more suitable for clinical events than for other types of outcomes (e.g., quality of life). Another limitation of using COPE is that the number of outcomes (and hence the number of trade-offs between outcomes) considered by guideline panels increases; more sophisticated economic modeling techniques will be needed to inform this process. This more sophisticated modeling can be thought of as a higher level synthesis, as it allows for synthesizing evidence for multiple outcomes (e.g., use of quality-adjusted life year [QALY]) and associated resource use (costs). This modeling activity can be proportionate to the resources available to conduct this work for a decision and the circumstance, with COPE being the fastest and most frugal approach.

4.2. Scenario 2: limited source of cost-effectiveness evidence

Sometimes, evidence on resource use and cost-effectiveness is limited and may come from a single source. In 2020, the World Health Organization developed guidelines for the management and care of drug-resistant tuberculosis [23]. For multidrug-resistant or rifampicin-resistant tuberculosis, the guidelines considered whether all-oral bedaquiline-containing shorter regimen of 9−12 months compared with standard care of shorter regimen with injectable agent recommended by the World Health Organization safely improve patient outcomes [23]. In this guideline, one CEA was identified, critically appraised, and included [24]. This study used a Markov cohort model to compare cost-effectiveness of the shorter (6−9 months) all-oral regimen containing bedaquiline, pretomanid, and linezolid (BPaL) compared with context-specific standard care for patients with extensively drug-resistant tuberculosis in South Africa, Georgia, and the Philippines [24]. Clinical effectiveness was derived from a single nonrandomized study [24]. Costs were estimated based on the literature and local consultation and converted to 2018 US dollars [24]. The outcome was measured using disability-adjusted life year (DALY) averted, representing the number of years of healthy life made possible by a given intervention. The analysis was conducted with a lifetime horizon from a health sector's perspective. Both deterministic and probabilistic sensitivity analyses were conducted to assess the uncertainty of CEA [24].

We created an economic evidence profile to summarize the principal findings and the certainty of CEA from the only available evaluation (Table 2). The cost of BPaL varied from $245 in the Philippines to $362 in South Africa, whereas the cost of standard care ranged from $450 to $624 at the intensive phase and from $88 to $239 at the continuation phase. The saving in total cost and DALYs averted by BPaL compared with standard care ranged from $1,410 and 8.73 in South Africa to $4,059 and 10.14 in Georgia (i.e., in both settings, BPaL was more effective and less costly and hence based on these data dominant).

The EtD framework summarizes the key design and the main findings of the CEA and the recommendations of the guideline panel (Table 3) [23]. The panel considered the cost saving by the 9−12-month all-oral regimen was moderate compared with the short injectable-medication containing regimen. Some key factors related to resource uses were noted, including existing stock of second-line medications during the transition, diagnostic capacity, and drug safety monitoring and management capacity. The magnitude of saving was dependent on the context, where the extent to which higher drug costs were outweighed by

Table 2. Economic evidence profile: BPaL vs. standard of care for XDR-TB

Study	BPal Cost/mo	Standard of care Cost/mo	Incremental cost per patient	Incremental effect per patient	ICER	Uncertainty	Certainty
One cost-effectiveness analysis [24]	South Africa $362	South Africa $624 (intensive phase) $239 (continuation phase)	−$1,410	8.73 DALYs averted	SoC dominated	Across all three settings, the finding that BPal was cost saving and more effective was robust in deterministic and probabilistic sensitivity analyses. The only exception was when the drug prices for BPaL became extremely expensive.	⊕⊕⊕○ Moderate
	Georgia $245	Georgia $450 (intensive phase) $89 (continuation phase)	−$4,059	10.14 DALYs averted	SoC dominated		
	The Philippines $245	The Philippines $455 (intensive phase) $88 (continuation phase)	−$3,864	9.93 DALYs averted	SoC dominated		

Abbreviations: BPaL, bedaquiline, pretomanid, and linezolid; XDR-TB, extensively drug-resistant tuberculosis; Soc, standard of care; DALY, disability-adjusted life yr.

the reduced costs of delivering injectable agents and treating recurrent and secondary cases may vary. The certainty of evidence was considered moderate as the estimates on resource use were similar across settings.

For cost-effectiveness, BPaL was dominant compared with standard care across all three settings. The cost-effectiveness results were robust in most sensitivity analyses, except when the drug prices became extremely expensive. Therefore, the panel considered that the cost-effectiveness evidence probably favors the shorter all-oral regimen over the injectable-containing regimens for this patient population.

The panel's final recommendation was that a shorter, all-oral, bedaquiline-containing regimen of 9−12 months' duration be used in eligible patients with confirmed multidrug-resistant or rifampicin-resistant tuberculosis, who have not been exposed to treatment with second-line tuberculosis medicines used in this regimen for more than 1 month, and in whom resistance to fluoroquinolones has been excluded. The conditionality of the recommendation was mainly attributed to the very low certainty in the efficacy evidence and the diagnostic capacity requirement [23].

4.3. Scenario 3: multiple sources of cost-effectiveness evidence

The number of publications on CEAs has been growing over the last decades [25]. Guideline developers may have to consider multiple CEAs when formulating recommendations. In a recommendation, from the European Commission Initiative Breast Cancer Guidelines, on whether an organized mammography screening program (MSP) or an opportunistic MSP (OS) should be used for early diagnosis of breast cancer in asymptomatic women [26−28], three relevant CEAs for the European setting were identified [29−31]. The economic evidence profile (Table 4) shows that in the two analyses conducted in Switzerland, one reported that the MSP was associated with €1,473 higher cost but gained 0.02 life-year compared with OS (ICER = €75,602) [29], whereas the other found that MSP gained more life years at lower costs (i.e., "dominant") [30]. The third CEA also reported that MSP was dominant to OS in terms of cost and life years in Austria [31]. Within these studies, the cost-effectiveness findings were robust in most sensitivity analyses (Table 4).

In the EtD frameworks (Table 5), the guideline panel highlighted the differences in the cost of MSP and OS, as well as ICER among the published CEAs [26,28]. Different models (i.e., Markov cohort model vs. microsimulation model) with the cost and effect data input from different time periods may contribute to the discrepancies. The participation rate of the screening program was also found to affect the cost-effectiveness estimates. There was consensus in the panel that the resources required for implementing the screening program would vary across countries. The panel noted that the MSP was associated with higher benefits, with the incremental cost varying across countries. As a result, the cost-effectiveness probably favors MSP.

The guidelines panel recommended using an organized screening program with MSP for early detection of breast cancer in asymptomatic women. This was a strong recommendation, with low certainty in the evidence on resource uses. The panel also noted that additional research on the cost-effectiveness of organized screening in different settings was needed [26,28].

5. Discussion

GRADE recognizes that evidence for relative efficiency of intervention alternatives can be useful in moving from evidence to recommendations and suggests including this

Table 3. Evidence to decision summary: resources required and cost-effectiveness all-oral shorter regimen containing bedaquiline vs. standard care for multidrug-resistant or rifampicin-resistant tuberculosis [23]

Resources required how large are the resource requirements (costs)?	
Judgment	**Research evidence**
• moderate savings	Compared with the injectable-containing shorter regimen, the 9–12-mo all-oral regimen is projected to be both cost-saving and more effective than the injectable-containing shorter regimen under nearly all modeled conditions. The projected cost savings average US $1,000 (2019) in South Africa and depend primarily on the extent to which higher drug costs are outweighed by the reduced costs of delivering injectable agents and treating recurrent and secondary cases (about US $2,000 per patient is saved in settings with two-times higher healthcare costs vs. <US $100 saved in settings with high drug costs but low healthcare costs).
Certainty of evidence of required resources: What is the certainty of the evidence of resource requirements (costs)?	
Judgment	**Research evidence**
• moderate	The cost savings relative to a short injectable-containing regimen are robust, except at extremes with drug costs, healthcare costs, or the cost of bedaquiline and other companion drugs.
Cost effectiveness: Does the cost-effectiveness of the intervention favor the intervention or the comparison?	
Judgment	**Research evidence**
• Probably favors the intervention	A cost-effectiveness model was developed, incorporating estimated health system costs with drug procurement, healthcare delivery, adverse events, retreatments, secondary cases, and morbidity and mortality associated with TB mortality and recurrence, treatment duration, drug toxicity, and TB transmission (ref). Costs and cost-effectiveness were evaluated in South Africa, with sensitivity analyses representing a range of drug and healthcare costs, high and low HIV coprevalence, 95% CIs for estimates of relative efficacy derived from statistical analysis of patient cohort data, and uncertainty in the values of parameters representing the natural history of TB. Compared with the injectable-containing shorter regimen, the 9–12-mo all-oral regimen is projected to be both cost saving and more effective than the injectable-containing shorter regimen under nearly all modeled conditions. The projected cost savings average US $1,000 (2019) in South Africa and depend primarily on the extent to which higher drug costs are outweighed by reduced costs of delivering injectable agents and treating recurrent and secondary cases (about US $2,000 saved per patient is saved in settings with two-times higher healthcare costs vs. < US $100 saved in settings with high drug costs but low healthcare costs). In a scenario where the all-oral regimen was no longer cost-saving because bedaquiline prices were four-fold higher than current pricing in South Africa or via the GDF, the incremental cost-effectiveness ratio of the all-oral regimen was US $400 per disability adjusted life-yr averted (range: US $100–US $900 across 95% CI for relative regimen efficacy).

information in GRADE EtDs. Some guideline developers have adapted GRADE evidence profiles and SoF tables to include summaries of the quality and principal findings of CEAs [32]. In this GRADE guideline article, we created economic evidence profiles that summarize key information about cost-effectiveness, including the cost of intervention, the cost of comparator, the incremental cost, the incremental effect, ICER, uncertainty from deterministic or probabilistic sensitivity analysis, and overall certainty of evidence. The economic evidence profile provides relevant information for the EtD process, documentation of the judgments decision-makers have to make, and informs recommendations.

5.1. Strengths and limitations

Note that the evidence profile does not substitute for a detailed description and appraisal of methods and results of each included CEA. Guidelines often lack appraisals of the certainty of clinical benefits and harms (models inputs) [33] and the certainty of CEA evidence [34]. Therefore, before incorporating economic evidence into the EtD frameworks, we recommend appraisal of reporting and methodologic quality of individual CEA that includes the assessment of certainty of all clinical input variables [6,35–37]. We also recommend transparent appraisal of the certainty of evidence from all relevant CEAs using GRADE domain of the directness, risk of bias in the body of evidence, heterogeneity in incremental cost per QALY estimates, and the publication bias [6]. Countries are increasingly developing their own "reference cases" for economic evaluation to guide the development and interpretation of CEAs for use in decision-making [38,39]. These reference cases would shape the assessment of relevance and quality of economic evidence in specific contexts.

There are challenges in summarizing economic evidence in a clear but concise way. The evidence profile examples presented in this article focus on pairwise comparisons. However, economic evaluations often compare multiple treatments. To present an ICER for each pairwise comparison in a series of economic evidence profiles, each of which accompanies the corresponding GRADE evidence

Table 4. Economic evidence profile: organized mammography screening vs. an opportunistic or nonorganized mammography screening program for breast cancer in asymptotic women

Study	Organized mammography screening	Opportunistic or nonorganized mammography screening program	Incremental cost per woman	Incremental effect per woman	ICER	Uncertainty	Certainty
	Cost/woman	Cost/woman					
Three CEAs [29–31]	Switzerland [29] €4,149	Switzerland [29] €2,676	€1,473	0.02 life-yr	€75,602 per life yr gained	ICER was sensitive to the mortality and incidence of breast cancer, cost of biopsy, and the sensitivity of MSP, as well as to the participant rate. The distribution of ICER ranging from $45,000 to $135,000, with highest probability of occurrence at $75,000.	⊕⊕⊕○ Moderate
	Switzerland [30] €1,630	Switzerland [30] €1,637	−€7	0.0171 life-yr 0.0159 QALY	Dominant	Different participation rates for the screening program were compared. In sensitivity analyses with different levels of false-negative rate, the cost-effectiveness finding was consistent across the scenarios.	
	Austria [31] €1,667	Austria [31] €1,678	−€11	0.009 life-yr	Dominant	MSP remained dominant in most deterministic sensitivity analyses. In probabilistic sensitivity analysis, the probability of MSP being cost-effective was 70% at the threshold of €50,000.	

Abbreviations: CEA, cost-effectiveness analysis; ICER, incremental cost-effectiveness ratio; MSP, organized mammography screening program; QALY, quality-adjusted life yr.

profile, it will not always be appropriate or useful to disaggregate the results of an economic evaluation. In an incremental analysis, each option is compared with the next most expensive, nondominated option. Not all ICERs are comparable. Many decision-makers recommend the use of generic outcome measure (e.g., QALY or DALY derived using very specific methods) in economic evaluations to allow for broad comparisons [40,41]. Nevertheless, there are some other sources of variability that make the ICER not comparable such as differences in the perspective of the analysis and the time horizon used, among others.

Another challenge arises when clinical or economic findings vary across patient groups. If relative treatment effects for clinical outcomes are inconsistent across different patient subgroups, the findings for each patient subgroup may be presented in separate rows of a GRADE evidence profile, each accompanied by an economic evidence profile. However, if the relative treatment effects for clinical outcomes are constant across subgroups, but the economic findings differ across subgroups (e.g., if incremental cost-effectiveness varies with baseline risk), it may be preferable to present a single GRADE evidence profile and an associated economic evidence profile with separate rows for each patient subgroup. Or, if economic findings vary across a continuum of risk, it may be simpler to present a threshold at which treatment becomes cost-effective. We now are working with the GRADE project group on economic evidence on an integration of these types of evidence profiles and summary of findings tables in GRADE's application GRADEpro to facilitate their development.

Although the number of economic evaluations has been increasing, cost-effectiveness evidence may not always be available for guideline developers, especially for innovative new treatments. Calculations using the COPE could offer a quick assessment on cost-effectiveness of alternative treatment strategies. However, this method should not replace the need for economic modeling when substantial resource use is expected throughout the course of treatment, and the comparison involves trade-offs among multiple important outcomes. Here, best practice would suggest the use of generic measures such as QALY or DALY [39].

Table 5. Evidence to decision summary: organized mammography screening vs. an opportunistic or nonorganized mammography screening program for breast cancer in asymptotic women [26]

Resources required how large are the resource requirements (costs)?	
Judgment	**Research evidence**
• Varies	The GDG reviewed evidence incorporated from three research studies [7,16,17], notes that the costs were lower for organized screening and higher in one study. The GDG discussed that the differences in the total costs of opportunistic vs. organized screening may be related to differences in the yr value of costs (one study uses 2004 cost value [16] and the other one uses 2007 cost value) [7], or it may also be related to the model inputs or type of modeling used (Markov modeling vs. microsimulation). For this reason, the quality of the evidence was downgraded to low. There was uncertainty in the results because of indirectness, information from two studies come from the same canton in Switzerland (Voud) and may not be able to be extrapolated [7,16]. The GDG noted that radiologist costs may be higher for opportunistic screening. The GDG also notes that in organized screening there may be additional administration costs; however, the cost per examination may be lower and would vary by country. The GDG notes that health-related costs may be higher if additional tests beyond mammogram are ordered as a result of opportunistic screening. The GDG also notes that the definition of organized screening may vary from country to country in Europe. The GDG agreed by consensus that the resources required would vary.
Certainty of evidence of required resources: What is the certainty of the evidence of resource requirements (costs)?	
Judgment	**Research evidence**
• Low	The quality is low due to imprecision and indirectness. Two studies reported organized screening as a dominant strategy that is more effective and less costly and the other one did not. The research was further downgraded for imprecision and indirectness. The GDG suggests that local data may be available in their own languages, and not published, to inform cost-effectiveness evidence. The GDG also notes that gray literature may also inform cost-effectiveness decision-making locally.
Cost effectiveness: Does the cost-effectiveness of the intervention favor the intervention or the comparison?	
Judgment	**Research evidence**
• Probably favors the intervention	The GDG reviewed evidence incorporated from three research studies [7,16,17]. Two studies demonstrated that organized screening was dominant; in the Neeser (2007) study, the ICER was 75,602 Euros per life yr gained. As one may expect that health benefits are higher in organized screening, costs in relation to these benefits will probably be favored. Costs may differ, but even if they are higher or lower, in most cases, organized screening will be cost-effective. The GDG agreed by consensus that the cost-effectiveness would probably favor the intervention in most settings.

There is rarely a simple relationship between research questions, a body of health evidence, an economic evaluation or a set of economic evaluations, and a set of recommendations. Although standard formats for economic evidence are useful, expressing those in complex evidence networks remains an area of research.

6. Conclusion

Cost-effectiveness evidence has played an increasingly important role in informing health-related recommendations. GRADE evidence profiles and SoF tables can be used to inform a quick assessment of cost-effectiveness of alternative treatments when there are no published economic evaluations. More often than not, there may be multiple relevant economic evaluations for which economic evidence profiles can provide top-level summaries of the quality and results of the evaluations. The evidence summary helps guideline panels to move from evidence to recommendation through the EtD frameworks. Further work is needed to address the presentational challenges of summarizing complex clinical and economic evidence networks in a clear but concise way and for the assessment of certainty in the evidence of modeled evidence [6].

Declaration of competing interest

The GRADE system has been developed by the GRADE Working Group. The named authors drafted and revised this article. F.X. is a member of the EuroQol Group and elected deputy chair of its Executive Committee. Seven authors of this article (I.S., L.V., F.R., M.F.D., J.L., K.H., and M.X.R.) are all convenors or members of the Campbell and Cochrane Economics Methods Group, a Cochrane Methods Group. F.X., I.S., L.V., Y.Z., M.X.R., C.C.A., P.A.C., and H.J.S. are members of the GRADE working group. The views expressed in this article are those of the authors and not necessarily those of any organization.

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